

Write or stick your
student number here



2025 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

General instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Student Number at the top of this page and at the top of each writing booklet

Total marks:
70

Section I – 10 marks (pages 2–7)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 8–13)

- Attempt Questions 11–15
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1** The polynomial equation $2x^3 - 3x^2 - 4x + 6 = 0$ has roots α , β and γ .

What is the value of $\alpha\beta\gamma$?

- A. -6
- B. -3
- C. -2
- D. $\frac{3}{2}$

- 2** The polynomial $P(x) = x^4 + ax^2 + b$ leaves a remainder of 3 when divided by $(x - 2)$.

What is the remainder when $P(x)$ is divided by $(x + 2)$?

- A. -3
- B. -1
- C. 3
- D. 7

- 3** In a barrel there are 50 marbles of various colours. Of these, 4 are green, 19 are blue, 11 are yellow, 13 are purple and 3 are orange.

What is the least number of marbles that can be selected from the barrel to ensure that 7 of the selected marbles are of the same colour?

- A. 26
- B. 30
- C. 31
- D. 35

4 Which of the following is the primitive of $\frac{1}{\sqrt{4-9x^2}}$?

A. $\frac{1}{6} \sin^{-1}\left(\frac{2x}{3}\right) + c$

B. $\frac{1}{3} \sin^{-1}\left(\frac{2x}{3}\right) + c$

C. $\frac{1}{6} \sin^{-1}\left(\frac{3x}{2}\right) + c$

D. $\frac{1}{3} \sin^{-1}\left(\frac{3x}{2}\right) + c$

5 A curve is defined by the parametric equations below.

$$x = \sin \theta$$

$$y = \cos^2 \theta - 3$$

Which of the following represents the Cartesian equation of the curve?

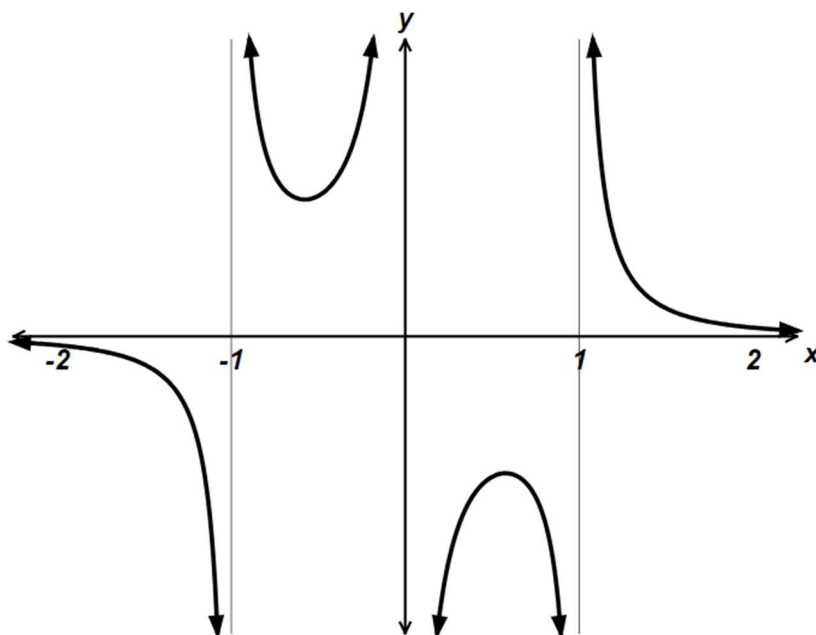
A. $y = x^2 - 3$

B. $y = -x - 2$

C. $y = -x^2 - 2$

D. $y = -2x^2 - 2$

- 6 The graph below shows $y = \frac{1}{f(x)}$.



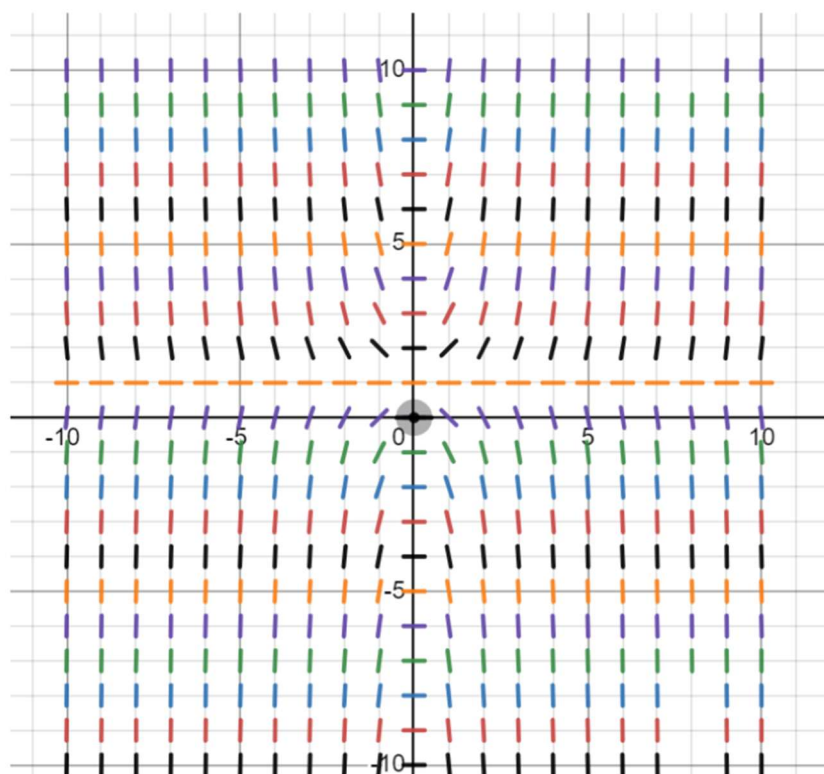
Which of the equations below could represent $y = f(x)$?

- A. $f(x) = x^2 - 1$
 - B. $f(x) = x(x^2 - 1)$
 - C. $f(x) = x^2(x^2 - 1)$
 - D. $f(x) = x^2(x^2 - 1)^2$
- 7 A particle's displacement is given by $x = 2\sin(\pi t) + 1$.

Which of the following statements is true?

- A. The maximum velocity of the particle occurs at $t = 2$ when $x = 1$.
- B. The maximum velocity of the particle occurs at $t = 2$ when $x = 2\pi$.
- C. The maximum velocity of the particle occurs at $t = 1.5$ when $x = -1$.
- D. The maximum velocity of the particle occurs at $t = 0.5$ when $x = 3$.

- 8 The diagram shows the direction field of a differential equation.

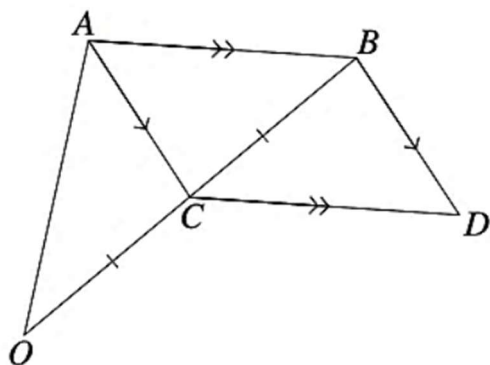


Which of the following differential equations could have the given direction field?

- A. $\frac{dy}{dx} = x^2 + y^2$
- B. $\frac{dy}{dx} = x^2 - y^2$
- C. $\frac{dy}{dx} = y^2 - x^2$
- D. $\frac{dy}{dx} = xy - x$

- 9 In the diagram, the position vectors of the points A and B are $\underline{a}$ and $\underline{b}$ respectively, where O is the origin.

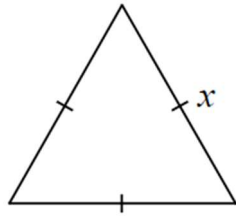
Point C is the midpoint of OB and point D is such that $ABDC$ is a parallelogram.



Which of the following is the position vector of D ?

- A. $\frac{3}{2}\underline{b} + \underline{a}$
B. $\frac{3}{2}\underline{b} - \underline{a}$
C. $\frac{1}{2}\underline{b} - \frac{1}{2}\underline{a}$
D. $\frac{1}{2}\underline{b} - \underline{a}$

- 10 The diagram shows an equilateral triangle with side length x cm and area A cm².



The area of the triangle is increasing at a rate of 6 cm² per second.

At what rate is the side length x expanding when the triangle has a side length of 4 cm?

- A. $\frac{2}{\sqrt{3}}$ cm/s
- B. $\sqrt{3}$ cm/s
- C. 6 cm/s
- D. $12\sqrt{3}$ cm/s

Section II

60 marks

Attempt Questions 11–15

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (12 marks) Use the Question 11 Writing Booklet

(a) (i) Express $2\cos x - \sin x$ in the form $R\cos(x + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 360^\circ$. 2

(ii) Hence find the range of the function $g(x) = (2\cos x - \sin x)^2$. 1

(b) Solve $-\frac{1}{x+1} \leq x+3$. 3

(c) A group of six girls and two boys are to be seated around a round table.

(i) In how many different ways can the eight people be seated? 1

(ii) What is the probability the two boys won't be seated next to each other? 2

(d) Use the substitution $x = 2\cos u$ to evaluate $\int_0^1 \frac{x^2}{\sqrt{4-x^2}} dx$. 3

End of Question 11

Question 12 (12 marks) Use the Question 12 Writing Booklet

- (a) (i) By letting $t = \tan \theta$, or otherwise, show that $\operatorname{cosec}(2\theta) - \cot(2\theta) = \tan \theta$. 2
- (ii) Hence find the exact value of $\tan \frac{\pi}{8}$. 1

- (b) Sketch the graph of $y = \pi - \cos^{-1}(2x)$, clearly indicating the endpoints and any intercepts with the coordinate axes. 2

- (c) Given that $y = e^{2x} + e^{-2x}$, determine the values of constants a and b that satisfy the differential equation $\frac{d^2y}{dx^2} + a \frac{dy}{dx} + by = 5e^{2x} + e^{-2x}$. 2

- (d) The population of a particular town is growing at a rate proportional to the excess of its current size N above 3000 individuals.

That is, N satisfies the equation $\frac{dN}{dt} = k(N - 3000)$.

At the start of 2020 the population was 25 000, and at the start of 2025 the population was 29 400.

- (i) Show that $N = 3000 + Ae^{kt}$ satisfies this model of population growth. 1
- (ii) Hence show that $A = 22\,000$ and $k = \frac{1}{5} \ln \left(\frac{6}{5} \right)$. 2
- (iii) In what year will the population first exceed 35 000? 1
- (iv) What will be the rate of population growth when the population reaches 35 000? 1

End of Question 12

Question 13 (12 marks) Use the Question 13 Writing Booklet

(a) (i) Find $\frac{d}{dx}(x \tan^{-1}x)$. **1**

(ii) Hence find $\int \tan^{-1}x \, dx$. **2**

- (b) A particle is fired from a point on the ground, taken to be the origin, with speed of projection 80 ms^{-1} at an angle α above the horizontal.

The particle needs to clear a fence 5 metres high at a horizontal distance 400 metres from the point of projection.

The position vector of the particle is $\underline{r}(t)$, where t is the time after projection. The acceleration due to gravity is $g = 10 \text{ ms}^{-2}$.

(i) Show that $\underline{r}(t) = (80t \cos \alpha) \underline{i} + (80t \sin \alpha - 5t^2) \underline{j}$. **3**

- (ii) Hence find the possible values of α for the particle to clear the fence, correct to the nearest minute. **3**

- (c) Use mathematical induction to prove that $9^{n+2} - 4^n$ is divisible by 5 for all positive integers n . **3**

End of Question 13

Question 14 (12 marks) Use the Question 14 Writing Booklet

- (a) A Maths club has $2n$ members, half of which are male and half female. 2

A Pi Day committee is made up of 3 members of this club.

By considering all different combinations for this committee, show that

$$n {}^nC_2 + {}^nC_3 = \frac{1}{2} {}^{2n}C_3, \text{ where } n \geq 3.$$

- (b) In the expansion of $(3 + 2x)^n$, the coefficients of x^2 and x are in the ratio 3:1. 3

Find the value of n .

- (c) Solve the differential equation $(x^2 + 1) \frac{dy}{dx} = 6xy$, given that $y = 2$ when $x = 1$. 3
Give your answer as the function y in terms of x .

- (d) An acute-angled triangle XYZ has an area of 40 square units. 4

The vector $\overrightarrow{YX} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$ and $\overrightarrow{YZ} = \begin{pmatrix} p \\ q \end{pmatrix}$.

Given $|\overrightarrow{YZ}| = 8\sqrt{5}$, find the possible values of p and q .

End of Question 14

Question 15 (12 marks) Use the Question 15 Writing Booklet

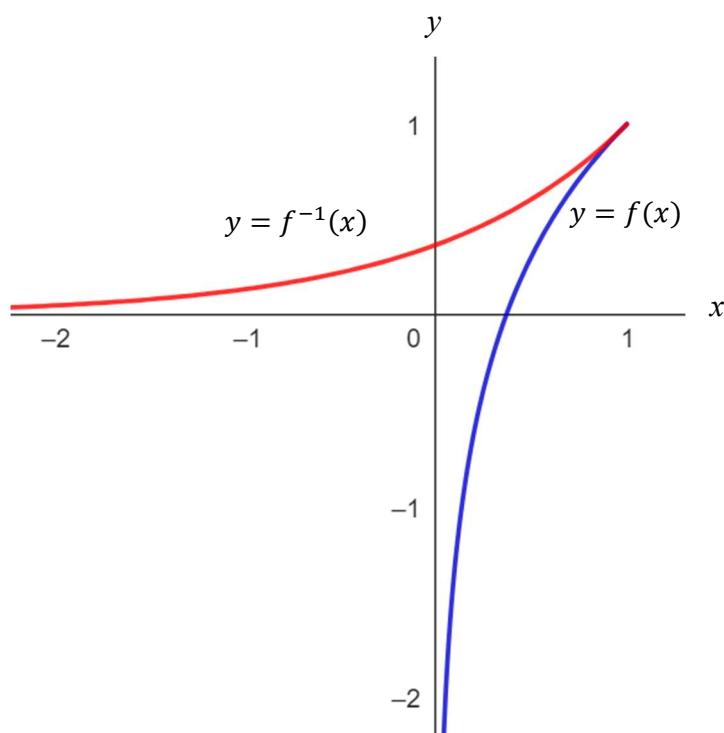
(a) (i) Show that $\sin(x-y)\sin(x+y) = \sin^2 x - \sin^2 y$. 2

(ii) Hence or otherwise, solve $\sin^2 3x - \sin^2 x = \sin 4x$ for $[0, \pi]$. 2

(b) The function $f(x) = 1 + \ln x$ is defined in the domain $(0, 1]$.

(i) Show that $\frac{d}{dx}(x \ln x) = 1 + \ln x$. 1

(ii) The diagram shows the graphs of the function $y = f(x)$ and the inverse function $y = f^{-1}(x)$ 3



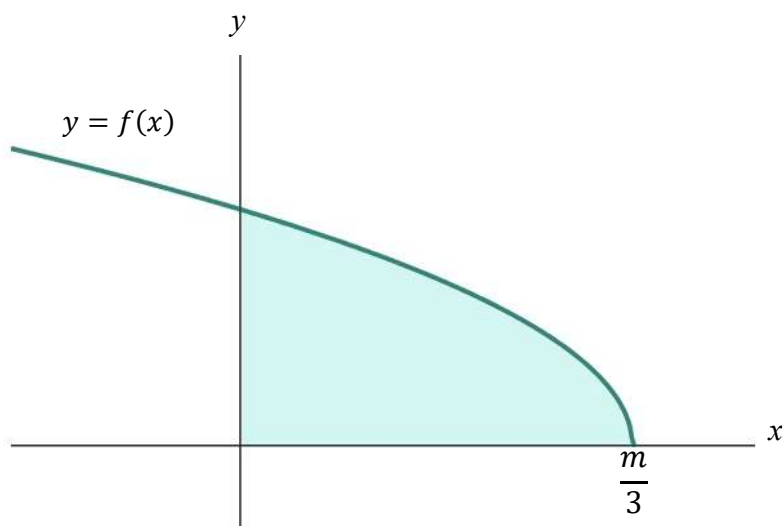
Find in simplest exact form the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = f^{-1}(x)$ and the coordinate axes.

Question 15 continues on page 13

Question 15 (continued)

- (c) Let $f(x) = \sqrt{m - 3x}$ for $x \leq \frac{m}{3}$. The graph of $y = f(x)$ is shown.

4



The shaded area, enclosed by the graph of $y = f(x)$, the x -axis and the y -axis, is rotated about the y -axis.

Find the value of m such that the volume of the solid formed is $\frac{5000\pi}{27}$ cubic units.

End of paper

Section I

Multiple choice answer key

Question	Answer
1	B
2	C
3	A
4	D
5	C
6	B
7	A
8	D
9	B
10	B

- 1 The polynomial equation $2x^3 - 3x^2 - 4x + 6 = 0$ has roots α , β and γ .

What is the value of $\alpha\beta\gamma$?

- A. -6
- B. -3
- C. -2
- D. $\frac{3}{2}$

Solution:

$$\begin{aligned}\alpha\beta\gamma &= -\frac{d}{a} \\ &= -\frac{6}{2} \\ &= -3\end{aligned}$$

$\rightarrow$ **B**

- 2 The polynomial $P(x) = x^4 + ax^2 + b$ leaves a remainder of 3 when divided by $(x - 2)$.

What is the remainder when $P(x)$ is divided by $(x + 2)$?

- A. -3
- B. -1
- C. 3
- D. 7

Solution:

$P(x)$ is even, so $P(-2) = P(2)$

Hence remainder = 3

$\rightarrow$ **C**

- 3 In a barrel there are 50 marbles of various colours. Of these, 4 are green, 19 are blue, 11 are yellow, 13 are purple and 3 are orange.

What is the least number of marbles that can be selected from the barrel to ensure that 7 of the selected marbles are of the same colour?

- A. 26
- B. 30
- C. 31
- D. 35

Solution:

Can have maximum of 4 green and 3 orange, then 6 each of blue, yellow or purple without having 7 of any one colour. Selecting one extra marble will ensure 7 of the same colour.

$$4 + 3 + 6 + 6 + 6 + 1 = 26$$

→ **A**

- 4 Which of the following is the primitive of $\frac{1}{\sqrt{4-9x^2}}$?

- A. $\frac{1}{6} \sin^{-1}\left(\frac{2x}{3}\right) + c$
- B. $\frac{1}{3} \sin^{-1}\left(\frac{2x}{3}\right) + c$
- C. $\frac{1}{6} \sin^{-1}\left(\frac{3x}{2}\right) + c$
- D. $\frac{1}{3} \sin^{-1}\left(\frac{3x}{2}\right) + c$

Solution:

Using reference sheet, $\frac{1}{\sqrt{4-9x^2}} dx = \frac{1}{3} \sin^{-1}\left(\frac{3x}{2}\right) + c$

→ **D**

- 5 A curve is defined by the parametric equations below.

$$x = \sin \theta$$

$$y = \cos^2 \theta - 3$$

Which of the following represents the Cartesian equation of the curve?

- A. $y = x^2 - 3$
- B. $y = -x - 2$
- C. $y = -x^2 - 2$
- D. $y = -2x^2 - 2$

Solution:

$$x = \sin \theta$$

$$x^2 = \sin^2 \theta$$

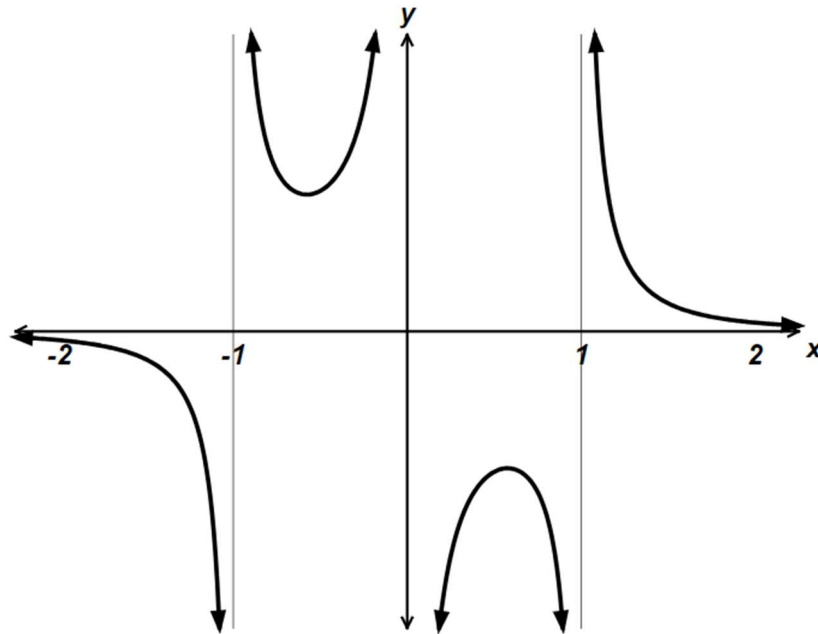
$$y = (1 - \sin^2 \theta) - 3$$

$$y = -\sin^2 \theta - 2$$

$$y = -x^2 - 2$$

$\rightarrow$ **C**

- 6 The graph below shows $y = \frac{1}{f(x)}$.



Which of the equations below could represent $y = f(x)$?

- A. $f(x) = x^2 - 1$
- B. $f(x) = x(x^2 - 1)$
- C. $f(x) = x^2(x^2 - 1)$
- D. $f(x) = x^2(x^2 - 1)^2$

Solution:

$f(x)$ has roots at $x = -1, 0, 1$

Change of sign at each root indicates they each have odd multiplicity

$$\begin{aligned} f(x) &= x(x-1)(x+1) \\ &= x(x^2-1) \end{aligned}$$

→ **B**

- 7 A particle's displacement is given by $x = 2\sin(\pi t) + 1$.

Which of the following statements is true?

- A. The maximum velocity of the particle occurs at $t = 2$ when $x = 1$.
- B. The maximum velocity of the particle occurs at $t = 2$ when $x = 2\pi$.
- C. The maximum velocity of the particle occurs at $t = 1.5$ when $x = -1$.
- D. The maximum velocity of the particle occurs at $t = 0.5$ when $x = 3$.

Solution:

$$x = 2\sin(\pi t) + 1$$

$$\dot{x} = 2\pi\cos(\pi t)$$

Maximum velocity occurs when $\cos(\pi t) = 1$

$$\pi t = 0, 2\pi$$

$$t = 0, 2$$

When $t = 0$, $x = 2\sin(0) + 1$

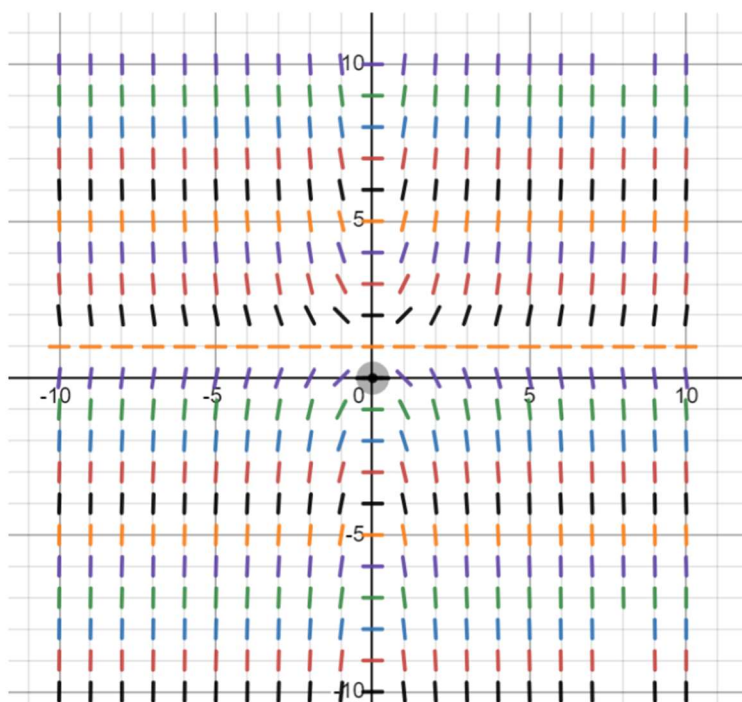
$$x = 1$$

When $t = 2$, $x = 2\sin(2\pi) + 1$

$$x = 1$$

$\rightarrow$ **A**

- 8 The diagram shows the direction field of a differential equation.



Which of the following differential equations could have the given direction field?

- A. $\frac{dy}{dx} = x^2 + y^2$
- B. $\frac{dy}{dx} = x^2 - y^2$
- C. $\frac{dy}{dx} = y^2 - x^2$
- D. $\frac{dy}{dx} = xy - x$

Solution:

When $y = 1$, $\frac{dy}{dx} = 0$

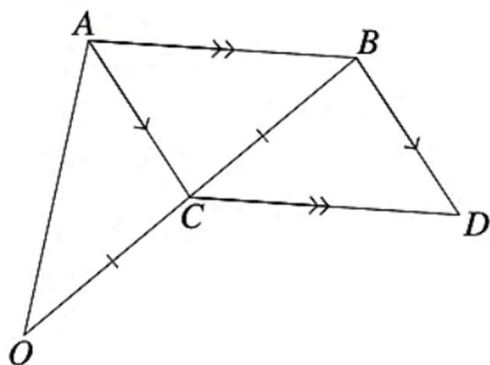
Consider option D: $\frac{dy}{dx} = x(y - 1)$

This is the only differential equation with a constant solution of $y = 1$

→ **D**

- 9 In the diagram, the position vectors of the points A and B are $\underline{a}$ and $\underline{b}$ respectively, where O is the origin.

Point C is the midpoint of OB and point D is such that $ABDC$ is a parallelogram.



Which of the following is the position vector of D ?

- A. $\frac{3}{2}\underline{b} + \underline{a}$
 B. $\frac{3}{2}\underline{b} - \underline{a}$
 C. $\frac{1}{2}\underline{b} - \frac{1}{2}\underline{a}$
 D. $\frac{1}{2}\underline{b} - \underline{a}$

Solution:

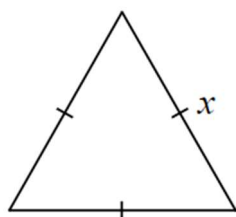
$$\begin{aligned}\overrightarrow{OC} &= \frac{1}{2} \overrightarrow{OB} \\ &= \frac{1}{2} \underline{b}\end{aligned}$$

$$\begin{aligned}\overrightarrow{CD} &= \overrightarrow{AB} \\ &= \underline{b} - \underline{a}\end{aligned}$$

$$\begin{aligned}\overrightarrow{OD} &= \overrightarrow{OC} + \overrightarrow{CD} \\ &= \frac{1}{2} \underline{b} + \underline{b} - \underline{a} \\ &= \frac{3}{2} \underline{b} - \underline{a}\end{aligned}$$

→ **B**

- 10 The diagram shows an equilateral triangle with side length x cm and area A cm².



The area of the triangle is increasing at a rate of 6 cm² per second.

At what rate is the side length x expanding when the triangle has a side length of 4 cm?

- A. $\frac{2}{\sqrt{3}}$ cm/s
- B. $\sqrt{3}$ cm/s
- C. 6 cm/s
- D. $12\sqrt{3}$ cm/s

Solution:

$$A = \frac{1}{2} \times x^2 \times \sin \frac{\pi}{3}$$
$$= \frac{\sqrt{3}}{4} x^2$$

$$\frac{dA}{dx} = \frac{\sqrt{3}}{2} x$$
$$= \frac{\sqrt{3}}{2} \times 4$$
$$= 2\sqrt{3}$$

$$\frac{dx}{dt} = \frac{dx}{dA} \times \frac{dA}{dt}$$
$$= \frac{1}{2\sqrt{3}} \times 6$$
$$= \sqrt{3}$$

→ **B**

Question 11

- (a) (i) Express $2\cos x - \sin x$ in the form $R\cos(x + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 360^\circ$. 2

Sample answer:

$$\begin{aligned} 2\cos x - \sin x &= R\cos(x + \alpha) \\ &= R(\cos x \cos \alpha - \sin x \sin \alpha) \end{aligned}$$

$$R\cos \alpha = 2$$

$$R\sin \alpha = 1$$

$$R^2 = 2^2 + 1^2$$

$$R = \sqrt{5}$$

$$\tan \alpha = \frac{1}{2}$$

$$\alpha = 26^\circ 34'$$

$$\therefore 2\cos x - \sin x = \sqrt{5}\cos(x + 26^\circ 34')$$

Criteria	Marks
Provides correct solution	2
Finds the value of R or α , or equivalent merit	1

- (ii) Hence find the range of the function $g(x) = (2\cos x - \sin x)^2$. 1

Sample answer:

$$\begin{aligned} g(x) &= \left[\sqrt{5}\cos(x + 26^\circ 34') \right]^2 \\ &= 5\cos^2(x + 26^\circ 34') \end{aligned}$$

$$\text{Range: } 0 \leq y \leq 5$$

Criteria	Marks
Provides correct answer	1

(b) Solve $-\frac{1}{x+1} \leq x+3$.

3

Sample answer:

Critical values:

$$x \neq -1$$

$$-\frac{1}{x+1} = x+3$$

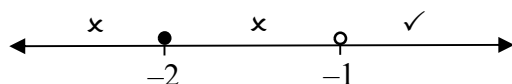
$$-1 = x^2 + 4x + 3$$

$$0 = x^2 + 4x + 4$$

$$0 = (x+2)^2$$

$$x = -2$$

Test values:



Hence $x = -2$ or $x > -1$

Criteria	Marks
Provides correct solution	3
Identifies the correct critical values, or equivalent merit	2
Excludes $x = -1$ OR finds $x = -2$ as a critical value, or equivalent merit	1

(c) A group of six girls and two boys are to be seated around a round table.

(i) In how many different ways can the eight people be seated?

1

Sample answer:

$$7! = 5040$$

Criteria	Marks
Provides correct answer	1

(ii) What is the probability the two boys won't be seated next to each other?

2

Sample answer:

Number of arrangements with boys seated together = $2 \times 6!$

$$= 1440$$

Number of arrangements boys not seated together = $5040 - 1440$

$$= 3600$$

$$P(\text{boys not seated together}) = \frac{3\,600}{5\,040}$$
$$= \frac{5}{7}$$

Criteria	Marks
Provides correct solution	2
Finds the number of arrangements with the boys seated next to each other, or equivalent merit	1

(d) Use the substitution $x = 2\cos u$ to evaluate $\int_0^1 \frac{x^2}{\sqrt{4-x^2}} dx$.

3

Sample answer:

$$\begin{array}{lll} x = 2\cos u & x = 0 & x = 1 \\ \frac{dx}{du} = -2\sin u & 2\cos u = 0 & 2\cos u = 1 \\ dx = -2\sin u \, du & u = \frac{\pi}{2} & u = \frac{\pi}{3} \end{array}$$

$$\begin{aligned} \int_0^1 \frac{x^2}{\sqrt{4-x^2}} dx &= \int_{\frac{\pi}{2}}^{\frac{\pi}{3}} \frac{4\cos^2 u}{\sqrt{4-4\cos^2 u}} \cdot (-2\sin u) \, du \\ &= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{8\cos^2 u \sin u}{2\sin u} \, du \\ &= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 4\cos^2 u \, du \\ &= 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 1 + \cos(2u) \, du \\ &= 2 \left[u + \frac{1}{2}\sin(2u) \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}} \\ &= 2 \left[\frac{\pi}{2} + \frac{1}{2}\sin\pi - \left(\frac{\pi}{3} + \frac{1}{2}\sin\frac{2\pi}{3} \right) \right] \\ &= 2 \left(\frac{\pi}{2} + 0 - \frac{\pi}{3} - \frac{\sqrt{3}}{4} \right) \\ &= \frac{\pi}{3} - \frac{\sqrt{3}}{2} \end{aligned}$$

Criteria	Marks
Provides correct solution	3
Obtains correctly simplified integrand in terms of $\cos u$, or equivalent merit	2
Finds an expression for dx in terms of u OR correctly adjust bounds of integration, or equivalent merit	1

Question 12

- (a) (i) By letting $t = \tan \theta$, or otherwise, show that $\operatorname{cosec}(2\theta) - \cot(2\theta) = \tan \theta$. 2

Sample answer:

Let $t = \tan \theta$

$$\begin{aligned}\operatorname{cosec}(2\theta) - \cot(2\theta) &= \frac{1+t^2}{2t} - \frac{1-t^2}{2t} \\ &= \frac{1+t^2-1+t^2}{2t} \\ &= \frac{2t^2}{2t} \\ &= t \\ &= \tan \theta\end{aligned}$$

Criteria	Marks
Provides correct solution	2
Correctly expresses the LHS in terms of t , or equivalent merit	1

- (ii) Hence find the exact value of $\tan \frac{\pi}{8}$. 1

Sample answer:

$$\begin{aligned}\tan \frac{\pi}{8} &= \operatorname{cosec} \frac{\pi}{4} - \cot \frac{\pi}{4} \\ &= \sqrt{2} - 1\end{aligned}$$

Criteria	Marks
Provides correct answer	1

- (b) Sketch the graph of $y = \pi - \cos^{-1}(2x)$, clearly indicating the endpoints and any intercepts with the coordinate axes. 2

Sample answer:

$y = \pi - \cos^{-1}(2x) \quad \rightarrow$

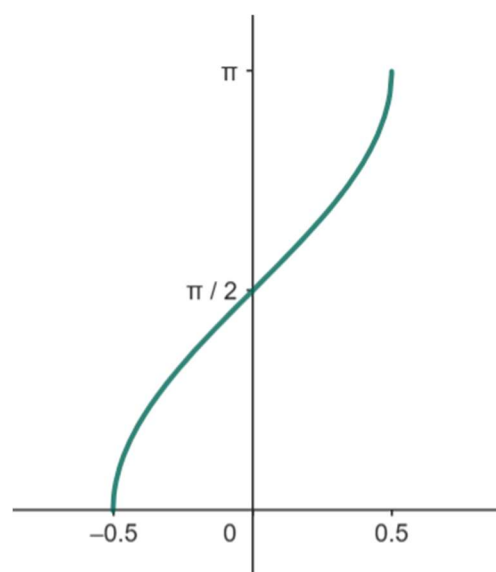
 horizontal dilation

 vertical reflection across x axis

 vertical translation up π units

Domain: $-1 \leq 2x \leq 1$
 $-\frac{1}{2} \leq x \leq \frac{1}{2}$

Range: $0 \leq \cos^{-1}(2x) \leq \pi$
 $-\pi \leq -\cos^{-1}(2x) \leq 0$
 $0 \leq \pi - \cos^{-1}(2x) \leq \pi$



Criteria	Marks
Provides correct graph	2
Correctly identifies domain OR range of the function OR correct shape	1

- (c) Given that $y = e^{2x} + e^{-2x}$, determine the values of constants a and b that satisfy the differential equation $\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 5e^{2x} + e^{-2x}$. 2

Sample answer:

$$y = e^{2x} + e^{-2x}$$

$$\frac{dy}{dx} = 2e^{2x} - 2e^{-2x}$$

$$\frac{d^2y}{dx^2} = 4e^{2x} + 4e^{-2x}$$

$$\begin{aligned}\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by &= 4e^{2x} + 4e^{-2x} + a(2e^{2x} - 2e^{-2x}) + b(e^{2x} + e^{-2x}) \\ &= (4 + 2a + b)e^{2x} + (4 - 2a + b)e^{-2x}\end{aligned}$$

$$4 + 2a + b = 5$$

$$2a + b = 1$$

$$4 - 2a + b = 1$$

$$-2a + b = -3$$

$$a = 1, b = -1$$

Criteria	Marks
Provides correct solution	2
Correctly finds first and second derivatives, or equivalent merit	1

- (d) The population of a particular town is growing at a rate proportional to the excess of its current size N above 3000 individuals.

That is, N satisfies the equation $\frac{dN}{dt} = k(N - 3000)$.

At the start of 2020 the population was 25 000, and at the start of 2025 the population was 29 400.

- (i) Show that $N = 3000 + Ae^{kt}$ satisfies this model of population growth.

1

Sample answer:

$$\begin{aligned} N &= 3000 + Ae^{kt} \\ \frac{dN}{dt} &= kAe^{kt} \\ &= k(3000 + Ae^{kt} - 3000) \\ &= k(N - 3000) \end{aligned}$$

Criteria	Marks
Provides correct solution	1

(ii) Hence show that $A = 22\,000$ and $k = \frac{1}{5} \ln\left(\frac{6}{5}\right)$.

2

Sample answer:

When $t = 0$, $N = 25\,000$

$$25000 = 3000 + Ae^0$$

$$A = 22000$$

When $t = 5$, $N = 29\,400$

$$29400 = 3000 + 22000 e^{5k}$$

$$26400 = 22000 e^{5k}$$

$$e^{5k} = \frac{6}{5}$$

$$5k = \ln\left(\frac{6}{5}\right)$$

$$k = \frac{1}{5} \ln\left(\frac{6}{5}\right)$$

Criteria	Marks
Provides correct solution	2
Correctly finds the value of A , or equivalent merit	1

(iii) In what year will the population first exceed 35 000?

1

Sample answer:

$$3000 + 22000 e^{kt} > 35000$$

$$22000 e^{kt} > 32000$$

$$e^{kt} > \frac{16}{11}$$

$$kt > \ln\left(\frac{16}{11}\right)$$

$$t > \frac{5 \ln\left(\frac{16}{11}\right)}{\ln\left(\frac{6}{5}\right)}$$

$$t > 10.27$$

Since $t = 10$ at the start of 2030, the population will first exceed 35 000 during 2030.

Criteria	Marks
Provides correct answer	1

- (iv) What will be the rate of population growth when the population reaches 35 000? **1**

Sample answer:

$$\begin{aligned}\frac{dN}{dt} &= k(N - 3000) \\ &= \frac{1}{5} \ln\left(\frac{6}{5}\right) \cdot (35000 - 3000) \\ &= 1166.857 \dots \\ &\approx 1167\end{aligned}$$

Criteria	Marks
Provides correct answer	1

Question 13

- (a) (i) Find $\frac{d}{dx}(x \tan^{-1}x)$. 1

Sample answer:

$$\begin{aligned}\frac{d}{dx}(x \tan^{-1}x) &= \tan^{-1}x \times 1 + x \times \frac{1}{1+x^2} \\ &= \tan^{-1}x + \frac{x}{1+x^2}\end{aligned}$$

Criteria	Marks
Provides correct answer	1

- (ii) Hence find $\int \tan^{-1}x \, dx$. 2

Sample answer:

$$\begin{aligned}\int \tan^{-1}x \, dx &= x \tan^{-1}x - \int \frac{x}{1+x^2} \, dx \\ &= x \tan^{-1}x - \frac{1}{2} \int \frac{2x}{1+x^2} \, dx \\ &= x \tan^{-1}x - \frac{1}{2} \ln(1+x^2) + c\end{aligned}$$

Criteria	Marks
Provides correct solution	2
Uses the result from (i) to find an expression for the required integral	1

- (b) A particle is fired from a point on the ground, taken to be the origin, with speed of projection 80 ms^{-1} at an angle α above the horizontal.

The particle needs to clear a fence 5 metres high at a horizontal distance 400 metres from the point of projection.

The position vector of the particle is $\underline{r}(t)$, where t is the time after projection. The acceleration due to gravity is $g = 10 \text{ ms}^{-2}$.

(i) Show that $\underline{r}(t) = (80t\cos\alpha)\underline{i} + (80t\sin\alpha - 5t^2)\underline{j}$.

3

Sample answer:

Horizontal components:

$$\ddot{x} = 0$$

$$\begin{aligned}\dot{x} &= \int 0 \, dt \\ &= c\end{aligned}$$

$$\text{When } t = 0, \dot{x} = 80\cos\alpha$$

$$\therefore \dot{x} = 80\cos\alpha$$

$$\begin{aligned}x &= \int 80\cos\alpha \, dt \\ &= 80t\cos\alpha + c\end{aligned}$$

$$\text{When } t = 0, x = 0$$

$$\therefore x = 80t\cos\alpha$$

Vertical components:

$$\ddot{y} = -10$$

$$\begin{aligned}\dot{y} &= \int -10 \, dt \\ &= -10t + c\end{aligned}$$

$$\text{When } t = 0, \dot{y} = 80\sin\alpha$$

$$\therefore \dot{y} = -10t + 80\sin\alpha$$

$$\begin{aligned}y &= \int -10t + 80\sin\alpha \, dt \\ &= -5t^2 + 80t\sin\alpha + c\end{aligned}$$

$$\text{When } t = 0, y = 0$$

$$\therefore y = -5t^2 + 80t\sin\alpha$$

$$\text{Hence } \underline{r}(t) = (80t\cos\alpha)\underline{i} + (80t\sin\alpha - 5t^2)\underline{j}$$

Criteria	Marks
Provides correct solution	3
Correctly finds an expression for the horizontal OR vertical component of $\underline{r}$, or equivalent merit	2
Makes some progress towards finding an expression for the horizontal OR vertical component of $\underline{r}$, or equivalent merit	1

- (ii) Hence find the possible values of α for the particle to clear the fence, correct to the nearest minute. 3

Sample answer:

When $x = 400$, we need $y > 5$

$$400 = 80t \cos \alpha$$

$$t = 5 \sec \alpha$$

$$-5(5 \sec \alpha)^2 + 80(5 \sec \alpha) \sin \alpha > 5$$

$$-125 \sec^2 \alpha + 400 \tan \alpha - 5 > 0$$

$$-125(1 + \tan^2 \alpha) + 400 \tan \alpha - 5 > 0$$

$$-125 \tan^2 \alpha + 400 \tan \alpha - 130 > 0$$

$$25 \tan^2 \alpha - 80 \tan \alpha + 26 < 0$$

Critical values:

$$\tan \alpha = \frac{80 \pm \sqrt{(-80)^2 - 4(25)(26)}}{2(25)}$$

$$= \frac{80 \pm \sqrt{3800}}{50}$$

$$\alpha = 70^\circ 33' \text{ or } 20^\circ 10'$$

Hence $20^\circ 10' < \alpha < 70^\circ 33'$

Criteria	Marks
Provides correct solution	3
Finds the required quadratic equation in terms of $\tan \alpha$ which will give the critical values	2
Finds the required expression for t in terms of α , or equivalent merit	1

- (c) Use mathematical induction to prove that $9^{n+2} - 4^n$ is divisible by 5 for all positive integers n . 3

Sample answer:

Step 1: Prove the statement to be true for $n = 1$

$$\begin{aligned} 9^3 - 4^1 &= 729 - 4 \\ &= 725 \\ &= 5 \times 145 \end{aligned}$$

$\therefore$ true for $n = 1$

Step 2: Assume the statement to be true for some positive integer $n = k$

$$\begin{aligned} \text{i.e. assume } 9^{k+2} - 4^k &= 5M \quad \text{for } M \in \mathbb{Z} \\ 9^{k+2} &= 5M + 4^k \end{aligned}$$

Step 3: Prove the statement to be true for $n = k + 1$, if it is true for $n = k$

$$\begin{aligned} 9^{k+3} - 4^{k+1} &= 9 \times 9^{k+2} - 4 \times 4^k \\ &= 9(5M + 4^k) - 4 \times 4^k \quad \text{using assumption} \\ &= 45M + 9 \times 4^k - 4 \times 4^k \\ &= 45M + 5 \times 4^k \\ &= 5(9M + 4^k) \\ &= 5J \quad \text{for } J \in \mathbb{Z} \end{aligned}$$

$\therefore$ true for $n = k + 1$, if true for $n = k$

Since true for $n = 1$, the statement is true for all $n \geq 1$ by mathematical induction.

Criteria	Marks
Provides correct proof	3
Proves the inductive step, or equivalent merit	2
Establishes the base case, or equivalent merit	1

Question 14

- (a) A Maths club has $2n$ members, half of which are male and half female. 2

A Pi Day committee is made up of 3 members of this club.

By considering all different combinations for this committee, show that

$$n {}^n\text{C}_2 + n {}^n\text{C}_3 = \frac{1}{2} 2^n {}^n\text{C}_3, \text{ where } n \geq 3.$$

Sample answer:

Total number of committees possible $= 2^n {}^n\text{C}_3$

Committees with 3 males, 0 females $= {}^n\text{C}_3$

Committees with 2 males, 1 female $= {}^n\text{C}_2 \times {}^n\text{C}_1$

Committees with 1 male, 2 females $= {}^n\text{C}_1 \times {}^n\text{C}_2$

Committees with 0 males, 3 females $= {}^n\text{C}_3$

Hence ${}^n\text{C}_3 + {}^n\text{C}_2 \times {}^n\text{C}_1 + {}^n\text{C}_1 \times {}^n\text{C}_2 + {}^n\text{C}_3 = 2^n {}^n\text{C}_3$

$$2 {}^n\text{C}_3 + 2 {}^n\text{C}_1 \times {}^n\text{C}_2 = 2^n {}^n\text{C}_3$$

But ${}^n\text{C}_1 = n$

So $2 {}^n\text{C}_3 + 2n {}^n\text{C}_2 = 2^n {}^n\text{C}_3$

$${}^n\text{C}_3 + n {}^n\text{C}_2 = \frac{1}{2} 2^n {}^n\text{C}_3$$

Criteria	Marks
Provides correct solution	2
Identifies total number of committees possible with different numbers of males, or equivalent merit	1

- (b) In the expansion of $(3 + 2x)^n$, the coefficients of x^2 and x are in the ratio 3:1.

3

Find the value of n .

Sample answer:

$$(3 + 2x)^n = {}^nC_0 3^n + {}^nC_1 3^{n-1}(2x) + {}^nC_2 3^{n-2}(2x)^2 + \dots$$

$$\text{Coefficient of } x^2 = {}^nC_2 \times 3^{n-2} \times 2^2$$

$$\text{Coefficient of } x = {}^nC_1 \times 3^{n-1} \times 2$$

$${}^nC_2 \times 3^{n-2} \times 2^2 = 3({}^nC_1 \times 3^{n-1} \times 2)$$

$${}^nC_2 \times 3^{n-2} \times 2^2 = {}^nC_1 \times 3^n \times 2$$

$${}^nC_2 \times 2 = {}^nC_1 \times 3^2$$

$$\frac{n!}{2!(n-2)!} \times 2 = n \times 9$$

$$n(n-1) = 9n$$

$$n^2 - n = 9n$$

$$n^2 - 10n = 0$$

$$n(n-10) = 0$$

$$n = 0 \text{ or } 10$$

$$\text{But } n > 0 \therefore n = 10$$

Criteria	Marks
Provides correct solution	3
Obtains a correct equation in terms of n , with factorials but without nC_k notation, or equivalent merit	2
Obtains a correct expression for one coefficient, or equivalent merit	1

- (c) Solve the differential equation $(x^2 + 1) \frac{dy}{dx} = 6xy$, given that $y = 2$ when $x = 1$. 3
- Give your answer as the function y in terms of x .

Sample answer:

$$\begin{aligned}
 (x^2 + 1) \frac{dy}{dx} &= 6xy \\
 \frac{1}{y} dy &= \frac{6x}{x^2 + 1} dx \\
 \int \frac{1}{y} dy &= \int \frac{6x}{x^2 + 1} dx \\
 \int \frac{1}{y} dy &= 3 \int \frac{2x}{x^2 + 1} dx \\
 \ln y &= 3 \ln(x^2 + 1) + c \\
 y &= e^{3 \ln(x^2 + 1) + c} \\
 y &= A e^{\ln(x^2 + 1)^3} \\
 y &= A(x^2 + 1)^3
 \end{aligned}$$

When $x = 1, y = 2$

$$\begin{aligned}
 2 &= A(1 + 1)^3 \\
 A &= \frac{1}{4} \\
 \therefore y &= \frac{1}{4}(x^2 + 1)^3
 \end{aligned}$$

Criteria	Marks
Provides correct solution	3
Correctly integrates to find an expression for y , without finding the constant of integration, or equivalent merit	2
Separates the variables in the differential equation, or equivalent merit	1

- (d) An acute-angled triangle XYZ has an area of 40 square units.

4

The vector $\overrightarrow{YX} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$ and $\overrightarrow{YZ} = \begin{pmatrix} p \\ q \end{pmatrix}$.

Given $|\overrightarrow{YZ}| = 8\sqrt{5}$, find the possible values of p and q .

Sample answer:

$$\begin{aligned} |\overrightarrow{YX}| &= \sqrt{6^2 + 2^2} \\ &= 2\sqrt{10} \end{aligned}$$

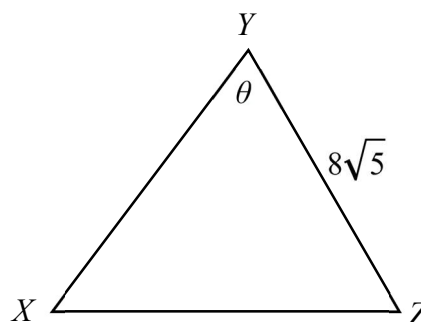
$$\text{Area} = \frac{1}{2} ab \sin C$$

$$40 = \frac{1}{2} \times 8\sqrt{5} \times 2\sqrt{10} \times \sin \theta$$

$$40 = 40\sqrt{2} \sin \theta$$

$$\sin \theta = \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4}$$



Consider $\overrightarrow{YX} \cdot \overrightarrow{YZ}$

$$6p + 2q = 2\sqrt{10} \times 8\sqrt{5} \times \cos \frac{\pi}{4}$$

$$6p + 2q = 80\sqrt{2} \times \frac{1}{\sqrt{2}}$$

$$6p + 2q = 80$$

$$q = 40 - 3p \quad (1)$$

Consider $|\overrightarrow{YZ}|$

$$\sqrt{p^2 + q^2} = 8\sqrt{5}$$

$$p^2 + q^2 = 320 \quad (2)$$

Substitute ① into ②

$$p^2 + (40 - 3p)^2 = 320$$

$$p^2 + 1600 - 240p + 9p^2 = 320$$

$$10p^2 - 240p + 1280 = 0$$

$$p^2 - 24p + 128 = 0$$

$$(p - 16)(p - 8) = 0$$

$$p = 16 \text{ or } 8$$

$$\begin{aligned} \text{When } p = 16, \quad q &= 40 - 3(16) \\ &= -8 \end{aligned}$$

$$\begin{aligned} \text{When } p = 8, \quad q &= 40 - 3(8) \\ &= 16 \end{aligned}$$

Hence either $p = 16$ and $q = -8$ or $p = 8$ and $q = 16$

Criteria	Marks
Provides correct solution	4
Obtains a pair of simultaneous equations relating p and q and correctly eliminates p or q , or equivalent merit	3
Obtains one equation relating p and q , or equivalent merit	2
Obtains a correct value for θ , or equivalent merit	1

Question 15

- (a) (i) Show that $\sin(x-y)\sin(x+y) = \sin^2x - \sin^2y$. 2

Sample answer:

$$\begin{aligned}\sin(x-y)\sin(x+y) &= (\sin x \cos y - \sin y \cos x)(\sin x \cos y + \sin y \cos x) \\ &= \sin^2x \cos^2y - \sin^2y \cos^2x \\ &= \sin^2x(1 - \sin^2y) - \sin^2y(1 - \sin^2x) \\ &= \sin^2x - \sin^2x \sin^2y - \sin^2y + \sin^2x \sin^2y \\ &= \sin^2x - \sin^2y\end{aligned}$$

Criteria	Marks
Provides correct solution	2
Makes significant progress towards solution e.g. eliminates compound angles AND expands brackets	1

- (ii) Hence or otherwise, solve $\sin^2 3x - \sin^2 x = \sin 4x$ for $[0, \pi]$. 2

Sample answer:

$$\begin{aligned}\sin^2 3x - \sin^2 x &= \sin 4x & 0 \leq x \leq \pi \\ \sin(3x-x) \sin(3x+x) &= \sin 4x & 0 \leq 2x \leq 2\pi \\ \sin 2x \sin 4x &= \sin 4x & 0 \leq 4x \leq 4\pi \\ \sin 2x \sin 4x - \sin 4x &= 0 \\ \sin 4x (\sin 2x - 1) &= 0\end{aligned}$$

$$\begin{aligned}\sin 4x &= 0 & \sin 2x &= 1 \\ 4x &= 0, \pi, 2\pi, 3\pi, 4\pi & 2x &= \frac{\pi}{2} \\ x &= 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi & x &= \frac{\pi}{4}\end{aligned}$$

Hence $x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$

Criteria	Marks
Provides correct solution	2
Obtains the equation $\sin 4x (\sin 2x - 1) = 0$, or equivalent merit	1

(b) The function $f(x) = 1 + \ln x$ is defined in the domain $(0, 1]$.

(i) Show that $\frac{d}{dx}(x \ln x) = 1 + \ln x$.

1

Sample answer:

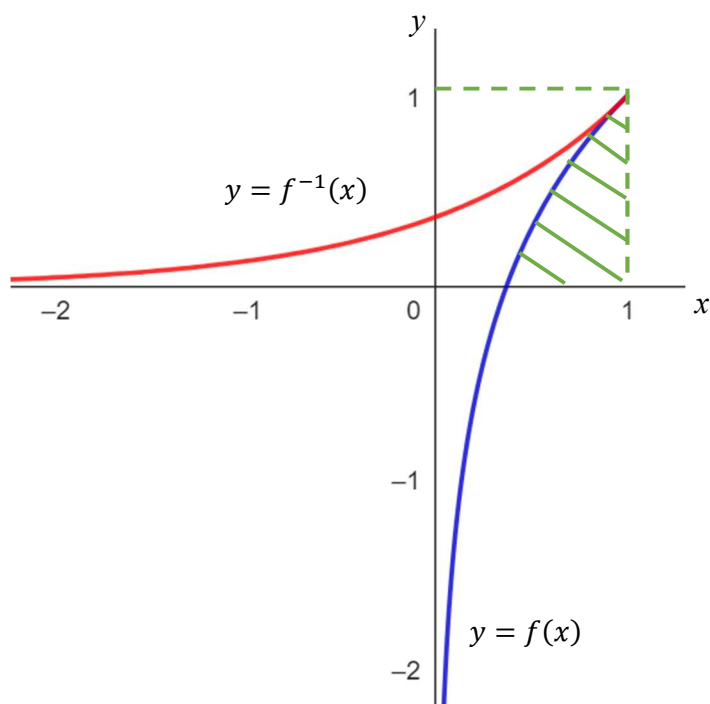
Using the product rule, $\frac{d}{dx}(uv) = vu' + uv'$

Let $u = x$, $v = \ln x$

$$\begin{aligned}\frac{d}{dx}(x \ln x) &= \ln x \times 1 + x \times \frac{1}{x} \\ &= \ln x + 1\end{aligned}$$

Criteria	Marks
Provides correct solution, demonstrating use of the product rule	1

- (ii) The diagram shows the graphs of the function $y=f(x)$ and the inverse function $y=f^{-1}(x)$. Find in simplest exact form the area of the region in the first quadrant bounded by the curves $y=f(x)$, $y=f^{-1}(x)$ and the coordinate axes. 3



Sample answer:

For x intercept of $y=f(x)$:

$$\begin{aligned} 1 + \ln x &= 0 \\ \ln x &= -1 \\ x &= \frac{1}{e} \end{aligned}$$

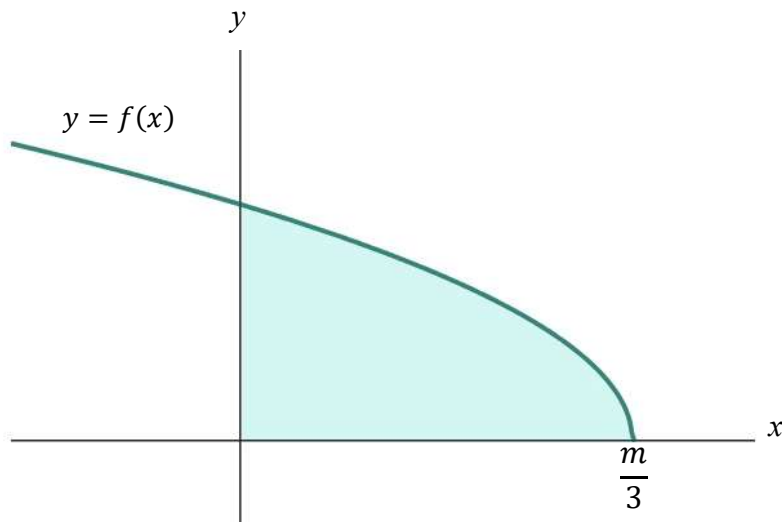
Required area = area of square – 2 x shaded area

$$\begin{aligned} \text{Area} &= 1 \times 1 - 2 \int_{\frac{1}{e}}^1 1 + \ln x \, dx \\ &= 1 - 2 \left[x \ln x \right]_{\frac{1}{e}}^1 \\ &= 1 - 2 \left[\ln(1) - \frac{1}{e} \ln\left(\frac{1}{e}\right) \right] \\ &= 1 - 2 \left(0 + \frac{1}{e} \right) \\ &= 1 - \frac{2}{e} \end{aligned}$$

Criteria	Marks
Provides correct solution	3
Obtains a correct integral expression for the required area	2
Finds the x intercept of $y=f(x)$, or equivalent merit	1

(c) Let $f(x) = \sqrt{m-3x}$ for $x \leq \frac{m}{3}$. The graph of $y=f(x)$ is shown.

4



The shaded area, enclosed by the graph of $y=f(x)$, the x -axis and the y -axis, is rotated about the y -axis.

Find the value of m such that the volume of the solid formed is $\frac{5000\pi}{27}$ cubic units.

Sample answer:

$$y = \sqrt{m-3x}$$

$$y^2 = m-3x$$

$$3x = m-y^2$$

$$x = \frac{1}{3}(m-y^2)$$

$$x^2 = \frac{1}{9}(m-y^2)^2$$

$$x^2 = \frac{1}{9}(m^2 - 2my^2 + y^4)$$

$$V = \pi \int_a^b x^2 dy$$

$$V = \frac{\pi}{9} \int_0^{\sqrt{m}} m^2 - 2my^2 + y^4 dy$$

$$\frac{5000\pi}{27} = \frac{\pi}{9} \left[m^2 y - \frac{2my^3}{3} + \frac{y^5}{5} \right]_0^{\sqrt{m}}$$

$$\frac{5000}{3} = m^2(\sqrt{m}) - \frac{2m}{3}(\sqrt{m})^3 + \frac{1}{5}(\sqrt{m})^5$$

$$\frac{5000}{3} = m^{\frac{5}{2}} - \frac{2}{3}m^{\frac{5}{2}} + \frac{1}{5}m^{\frac{5}{2}}$$

$$\frac{5000}{3} = \frac{8}{15}m^{\frac{5}{2}}$$

$$3125 = m^{\frac{5}{2}}$$

$$m = 3125^{\frac{2}{5}}$$

$$m = 25$$

Criteria	Marks
Provides correct solution	4
Obtains correct equation to be solved for m , or equivalent merit	3
Obtains correct integral expression for the required volume, or equivalent merit	2
Obtains correct expression for x^2 OR correct bounds of integration, or equivalent merit	1